

UI Demonstration & Screen Walkthrough Script

This script provides a step-by-step guide for the team member demonstrating the live user interface of the Human Development Index (HDI) Predictor application.

SCENE 1: Introduction & Landing Page (home.html)

- **Visual on Screen:** Browser open at `http://127.0.0.1:5000/`. Hovering cursor over the nav logo and hero badge.
- **Action:** Scroll down slowly to reveal the three dimension cards, then scroll back up.
- **Presenter Voiceover:**

"To begin our live demonstration, let's explore our landing page dashboard. We've built a modern, responsive interface using a premium dark gradient backdrop with glassmorphic cards.

>

On the main dashboard, users are introduced to the core pillars of the Human Development Index: a long and healthy life, education access, and GNI standard of living.

>

Next, let's navigate to our calculator form by clicking either the 'Predict' tab in the navigation bar or our main call-to-action button, 'Make a Prediction'."

SCENE 2: The Interactive Prediction Form (indexnew.html)

- **Visual on Screen:** Navigation transition to the `/predict` route. Cursor points to form inputs.
- **Action:** Hover over the range hints (e.g. `50-85 yrs`) next to each input field label.
- **Presenter Voiceover:**

"Here is our interactive parameter simulation dashboard.

>

The input form accepts country categories and the four key developmental indicators. To guarantee stable prediction results, we implemented native input boundaries on all fields:

- Life expectancy limits inputs between 50 and 85 years.
- Expected and Mean schooling years are validated for ranges 5 to 20, and 3 to 15.

- GNI per capita validates economic scales between \$500 and \$75,000.

>

Let's run a live prediction to see the system in action."*

SCENE 3: Scenario 1 — Highly Developed Nation Profile

- **Visual on Screen:** Inputting values for a highly developed country.
- **Action:** Enter the following values:
- Country: `Country\_A`
- Life Expectancy: `82.0`
- GNI per Capita: `55000`
- Expected Years of Schooling: `16.0`
- Mean Years of Schooling: `13.0`
- Click  **Predict HDI Score.**
- **Presenter Voiceover:**

"For our first scenario, we'll enter parameters representing a highly developed national profile.

>

As we submit, the backend instantly processes our inputs through our serialized model.

>

The result card displays a predicted HDI score of 1.0, automatically mapping it to the 'Very High' development tier with a green badge.

>

Below, you can see our visual progress gauge bar representing the country's development level, alongside targeted recommendations to sustain healthcare investments and promote technology sectors."*

SCENE 4: Scenario 3 — Intervention Planning (Low Development)

- **Visual on Screen:** Return to the predict form and enter low-development indicators.
- **Action:** Enter the following values:
- Country: `Country\_C`
- Life Expectancy: `52.0`
- GNI per Capita: `1500`

- Expected Years of Schooling: `5.0`
- Mean Years of Schooling: `3.0`
- Click 🗨️ **Predict HDI Score**.
- **Presenter Voiceover:**

*"Now, let's simulate a profile requiring development intervention.

>

Upon clicking predict, our score shifts to 0.5408, falling into the 'Low' development tier (marked by a red badge).

>

The progress gauge updates to reflect the low score, and the recommendations engine displays key areas requiring urgent intervention, such as basic vaccinations and primary education funding."*

SCENE 5: Model Information & Analytics (model_info.html)

- **Visual on Screen:** Click the **Model Info** tab in the navigation bar.
- **Action:** Scroll down slowly, pausing first at the mathematical formulation card, then at the feature weights table, and finally scrolling through the large plots gallery.
- **Presenter Voiceover:**

*"Lastly, let's explore our 'Model Info' tab, which serves as a detailed report on our system's performance.

>

At the top, we display our model KPI metrics, showing our trained Linear Regression model achieving an R-Squared accuracy score of 0.8834.

>

The mathematical layout shows our intercept and coefficients, with EYS and MYS having the highest positive impact on the index.

>

As we scroll down, we find our large, detailed Exploratory Data Analysis plots:

- The Correlation Heatmap shows feature correlations.
- The Schooling and Health Strip Plots map indicator distributions across the dataset.

>

This page provides a transparent overview of our model's performance and training metrics."*

SCENE 6: Conclusion

- **Visual on Screen:** Return to the landing page.
- **Presenter Voiceover:**

*"That concludes our live user interface walkthrough. The HDI Predictor successfully connects machine learning models with a modern, visual, and user-friendly interface.

>

Thank you for watching!"*